

Learning Action-Video Compatibility as a Reference-Free Verifier for Embodied World Models

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Abstract: World models allow embodied agents to imagine and compare possible futures before acting. However, when imagined rollout becomes inconsistent with the physical consequences, verification with ground-truth futures is difficult due to unavailable real-world resources. Rather than continually scaling or redesigning generative backbones, we ask whether a reference-free verifier can identify action-consistent futures without real-world execution. Inspired by CLIP’s success in contrastively aligning image-text pairs, we propose VA-CLIP, a simple yet effective video-action alignment framework that learns correspondences between action chunks and video transitions, which are inherently embedded in real robot interaction data. Specifically, VA-CLIP learns a reference-free action–video compatibility critic by contrastively aligning temporally matched visual transitions and motor action chunks in a shared L2-normalized space. Extensive experiments across simulated and real-robot datasets show that VA-CLIP learns an effective action–video compatibility space, achieving above 86.3% R@5 on video-to-action and action-to-video retrieval, above 96.7% accuracy in counterfactual discrimination, and a positive correlation with DINOv2 similarity. When used as a plug-and-play test-time verifier for world action models, VA-CLIP improves both IDM-based and Joint-based WAM formulations, increasing FastWAM-IDM success from 64.0% to 77.0% on eight challenging RoboTwin tasks and increasing FastWAM-Joint success from 42.5% to 77.5% in real-robot evaluations.

Keywords: World Action Models, Robot Manipulation, Contrastive Learning

1 Introduction

As generative models that predict future observations conditioned on candidate actions, world models [1] provide an internal imagination loop for embodied decision-making, enabling agents to assess behavioral consequences before execution for planning, policy learning, evaluation, and test-time improvement [2, 3, 4, 5, 6, 7, 8]. Recent advances further move from compact latent dynamics to video-centric, action-conditioned simulators that generate realistic multi-step rollouts [9, 10, 11, 12, 13, 14, 15, 16, 17]. In parallel, world-action models couple predictive dynamics with action learning, suggesting that imagined futures can provide decision-relevant signals rather than merely visual forecasts [18, 19, 20, 21, 22, 23, 24, 25].

However, world model imagination is only useful if it faithfully reflect the causal consequences of the conditioning actions [17]. In practice, the model must answer a counterfactual question without access to the ground-truth future: what would happen if this specific sequence of actions were executed from the current state? The model may produce a visually plausible but action-inconsistent video, delivering misleading signals to downstream decision-making. This creates a core need for a verifier that can judge the consistency of actions without a reference to the future.

Given the success of CLIP [26], we formulate reference-free verification as an *action–video compatibility* problem: given an action chunk and a candidate visual transition, the verifier scores whether

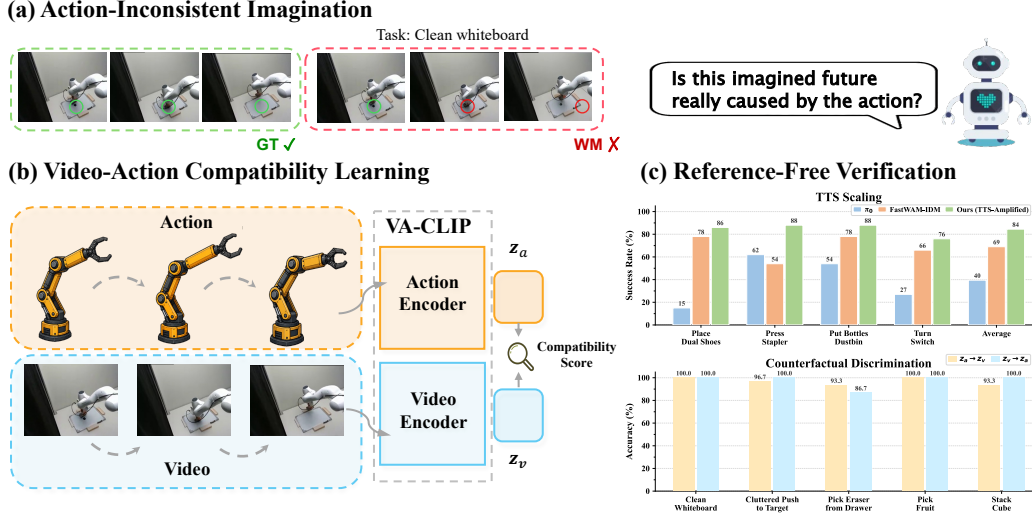


Figure 1: **Reference-free verification of action-consistent imagined rollouts.** (a) A world model may generate a visually plausible but action-inconsistent future for the same task. (b) VA-CLIP learns an action–video compatibility space by aligning conditioning action chunks with visual transition chunks. (c) The learned score enables reference-free verification, supporting both counterfactual discrimination and test-time rollout selection for world action models.

the transition is consistent with the action. Robot trajectories naturally provide supervision for this formulation, as temporally aligned action chunks and visual transitions form positive pairs. However, learning this compatibility is non-trivial because robot actions are low-dimensional continuous control signals, whereas videos are high-dimensional observations with appearance variation, temporal dynamics, and state-dependent physical effects.

Building on these insights and challenges, we introduce VA-CLIP, which adopts contrastive learning paradigm to learn a reference-free video-action compatibility critic. During training, we encode motor action chunks and their induced visual transitions into a shared, L2-normalized embedding space and optimize bidirectional InfoNCE loss [27, 26]. This objective encourages the model to capture action-effect correspondence, yielding a compatibility score for action-consistency verification.

We evaluate VA-CLIP across RoboTwin [28] simulation settings and self-collected Franka real-robot data. Across these settings, the learned compatibility score delivers strong bidirectional action–video retrieval and counterfactual discrimination, is sensitive to both action corruption and temporal misalignment, and correlates positively with DINOv2 [29] similarity between ground-truth and replayed rollouts. We further show that VA-CLIP can be used as a plug-and-play test-time verifier for world action models, improving success rate by test-time selection on both IDM-based and Joint-based WAM formulations; in particular, it improves the average success rate of FastWAM-IDM [22] by 13.0% on eight representative challenging RoboTwin tasks.

Our contributions can be summarized as:

- **Action-video compatibility learning.** We formulate reference-free verification as a cross-modal representation-learning problem over temporally aligned motor-action chunks and visual transitions, and learn the compatibility signal via bidirectional contrastive alignment.
- **A simple yet effective video-action alignment framework.** We introduce VA-CLIP, a lightweight CLIP-style framework that aligns continuous action chunks with visual transition chunks from paired robot trajectories, and applies to both simulated and real-robot interaction data.
- **Systematic validation and test-time application.** We show that VA-CLIP captures action-effect correspondence across simulated and real-robot manipulation data, responds to ac-

tion and temporal perturbations, ranks counterfactual transitions, correlates positively with DINOv2 similarity and improves world-action-model rollout selection at test time.

2 Related Works

2.1 Embodied World Models and World-Action Models.

World models allow agents to evaluate candidate behaviors through imagined rollouts rather than direct execution [1, 3, 4, 5]. This paradigm has been extended to robotic control, supporting planning and data-efficient decision making [6, 7, 30, 2, 8]. Recent work further moves to generalizable video-centric world models and world-action models [31, 9, 19, 13, 32, 18, 20, 23, 24, 25, 21, 22]. However, these methods mainly generate or exploit imagined futures, while VA-CLIP addresses a complementary problem: reference-free verification of whether a candidate rollout is compatible with its conditioning action sequence when no ground-truth future is available.

2.2 Multimodal Contrastive Learning for Robotics.

Contrastive learning has become a standard approach for aligning heterogeneous modalities, with CLIP-style dual encoders learning transferable representations by pulling matched image–text pairs together and pushing mismatched pairs apart [33, 34, 26, 35, 36]. Recent robotic studies further adapt contrastive alignment to action-centric settings. CLAP aligns video transitions with proprioceptive latent actions to extract executable action codes for VLA pretraining [37], while CoVer learns a verifier by vision–language–action alignment for test time scaling [38]. In contrast, VA-CLIP learns action–video compatibility for embodied world models, directly scoring whether a continuous action chunk is consistent with the visual transition it induces.

3 Method

3.1 VA-CLIP Architecture

VA-CLIP consists of two modality-specific encoders that map a visual transition chunk and its aligned action chunk into a shared normalized embedding space. For the visual branch, we first encode the sampled video chunk with a frozen WAN2.2 [39] VAE and treat the resulting spatio-temporal latents as visual tokens. These tokens are projected to the model dimension, augmented with positional embeddings, processed by a lightweight Transformer [40] encoder, and aggregated by learnable-query cross attention into a fixed-dimensional visual embedding. The action branch follows a parallel design: continuous action steps are projected into tokens, combined with positional embeddings, encoded by a Transformer, and pooled by learnable queries into an action embedding. Both embeddings are projected and ℓ_2 -normalized, and their inner product defines the action–video compatibility score:

$$s_\theta(v, u) = f_v(v)^\top f_u(u),$$

where $f_v(v)$ and $f_u(u)$ denote the normalized visual and action embeddings.

3.2 Dual-Granularity Contrastive Learning

VA-CLIP learns compatibility between temporally extended visual transitions and action sequences. To avoid tying the representation to a single temporal horizon, we construct two aligned views from each trajectory segment: a full-window view (v_t^H, u_t^H) and a half-window view $(v_t^{H/2}, u_t^{H/2})$. The half-window view uses the first half of the same temporal window, with the action chunk truncated to the aligned time range. Both views are encoded by the same video and action encoders and are included in the same contrastive batch. This dual-granularity construction augments action–transition supervision across temporal extents without requiring additional labels.

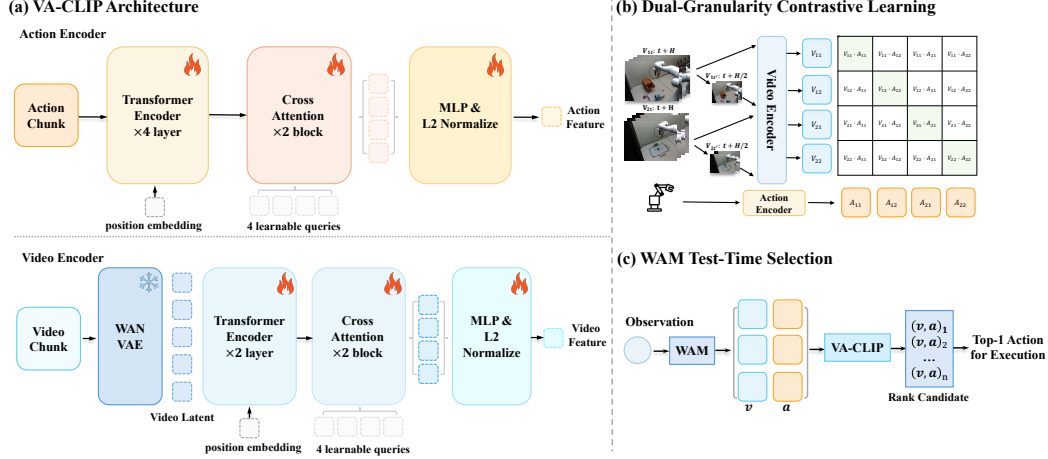


Figure 2: **Overview of VA-CLIP.** (a) VA-CLIP encodes continuous action chunks and visual transition chunks into a shared normalized embedding space. (b) Dual-granularity contrastive learning aligns full-horizon and prefix action–transition pairs from robot trajectories, pulling matched pairs together and separating mismatched pairs. (c) At test time, VA-CLIP serves as a plug-and-play verifier for world action models by scoring candidate rollouts with their conditioning actions and selecting the most action-consistent one for execution.

108 Let $\{(z_v^i, z_u^i)\}_{i=1}^M$ denote the normalized video and action embeddings after including both granu-
 109 larities. We compute pairwise action–video similarities with a learnable temperature scale γ :

$$\ell_{ij}^{v \rightarrow u} = \gamma(z_v^i)^\top z_u^j, \quad \ell_{ij}^{u \rightarrow v} = \gamma(z_u^i)^\top z_v^j.$$

110 Each embedding is matched with its temporally aligned counterpart as the positive pair, while all
 111 other embeddings in the batch serve as negatives. We optimize a symmetric contrastive objective:

$$\mathcal{L} = \frac{1}{2} [\text{CE}(\ell^{v \rightarrow u}, y) + \text{CE}(\ell^{u \rightarrow v}, y)], \quad y_i = i.$$

112 This objective aligns matched action–transition pairs and separates mismatched pairs, yielding a sin-
 113 gle compatibility space for reference-free action-consistency verification across temporal horizons.

114 3.3 Plug-and-Play Test-Time Selection

115 VA-CLIP can be used as a plug-and-play verifier for test-time rollout selection in world ac-
 116 tion models. Given the current observation, a WAM generates N candidate rollout-action pairs
 117 $\{(\hat{v}^{(n)}, \hat{u}^{(n)})\}_{n=1}^N$, where $\hat{v}^{(n)}$ is an imagined visual transition and $\hat{u}^{(n)}$ is the corresponding can-
 118 didate action sequence. These pairs may be obtained from different WAM formulations: in IDM
 119 models, actions are inferred from generated visual rollouts, while in Joint models, visual rollouts
 120 and actions are generated jointly.

121 We score each candidate pair with VA-CLIP:

$$q^{(n)} = s_\theta(\hat{v}^{(n)}, \hat{u}^{(n)}),$$

122 and select the candidate with the highest action–video compatibility:

$$(\hat{v}^*, \hat{u}^*) = (\hat{v}^{(n^*)}, \hat{u}^{(n^*)}), \quad n^* = \arg \max_n q^{(n)}.$$

123 The selected action sequence $\hat{u}^*$ is then executed by the robot. This procedure requires no ground-
 124 truth future observations and does not modify or retrain the underlying WAM, making VA-CLIP a
 125 lightweight test-time selector for more action-consistent imagined rollouts.



Figure 3: **Overview of the datasets used in our experiments.** We evaluate VA-CLIP across three complementary data regimes to demonstrate its applicability to diverse robot interaction settings.

Table 1: **Video-action retrieval results on validation sets.** We report bidirectional Top- k accuracy and the average VA-CLIP score of ground-truth action-video pairs to evaluate whether VA-CLIP learns a shared compatibility space between action chunks and visual transitions.

Dataset	Video-to-Action			Action-to-Video			GT Score
	Top-1	Top-5	Top-10	Top-1	Top-5	Top-10	
RoboTwin-Clean & Random	84.9	98.8	99.5	82.0	97.1	98.5	0.912
RoboTwin-Success & Fail	62.3	96.4	98.5	59.3	95.4	97.9	0.886
Franka	35.8	90.8	97.7	28.6	86.3	96.6	0.676

4 Experiment

4.1 Experimental Setup

Datasets. We train and evaluate VA-CLIP separately on three complementary datasets, as shown in Fig. 3. RoboTwin [28]-Clean & Random features appearance and object variations, RoboTwin-Success & Fail captures action-state diversity across execution outcomes, and Franka evaluates the framework under real-world dynamics with camera deviations.

Research Questions. We structure our evaluation around three questions.

- **RQ1:** Is VA-CLIP a feasible and interpretable action-video evaluator? We assess both retrieval accuracy and the factors that affect its scores.
- **RQ2:** Can VA-CLIP serve as a reference-free proxy for reference-based metrics when ground-truth trajectories are unavailable?
- **RQ3:** Can VA-CLIP improve world-action models by guiding test-time scaling?

4.2 Feasibility and Score Interpretability

Evaluation protocol. We evaluate whether VA-CLIP learns a meaningful action-video compatibility space from two perspectives. First, we measure bidirectional retrieval on the validation splits of the three datasets introduced in Sec. 4.1: RoboTwin Random-Clean ($N = 63,528$), RoboTwin Fail-Success ($N = 218,677$), and Franka real-robot data ($N = 33,590$). All splits are constructed at the trajectory level: trajectories in the validation set are never observed during training, and N denotes the total number of action-video chunks extracted from the held-out validation trajectories. For each dataset, we build a validation gallery of paired action chunks and video transitions, and report Top- k accuracy for both video-to-action and action-to-video retrieval using the learned compatibility score. Second, we probe score interpretability by evaluating how the compatibility score changes under controlled action perturbations and temporal shifts.

Results. Table 1 shows that VA-CLIP achieves strong bidirectional retrieval performance across all three validation sets. Top-1 accuracy ranges from 35.8% on Franka to 84.9% on RoboTwin-Clean, while Top-5 accuracy exceeds 86% in every setting, indicating that the learned space is

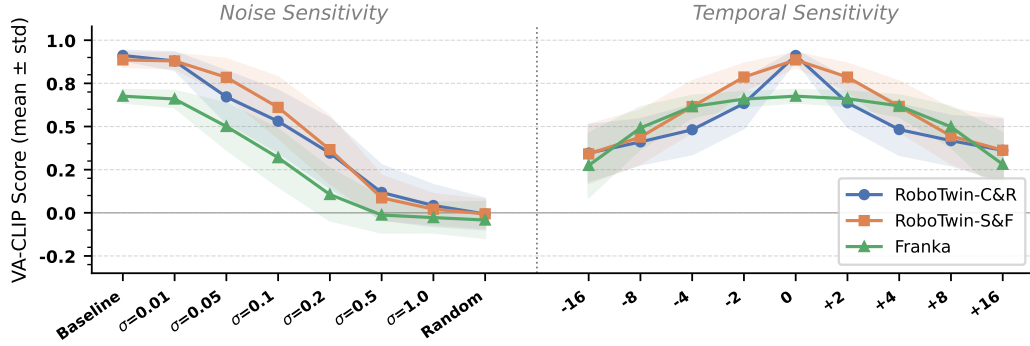


Figure 4: **Sensitivity analysis of VA-CLIP scores.** VA-CLIP is sensitive to noise injection and temporal misalignment, suggesting that its score is driven by action-conditioned transition dynamics.

Table 2: **Counterfactual discrimination results using cosine-of-means VA-CLIP scores.** A2V and V2A denote action-to-video and video-to-action three-way ranking accuracy, respectively. Pos. and CF denote the average scores of matched and counterfactual pairs. Gap measures their average separation, while Hard Marg. measures the margin against the strongest counterfactual alternative.

Task	A2V	V2A	Pos.	CF	Gap	Hard Marg.
Clean Whiteboard	100.0	100.0	0.466	0.287	0.178	0.138
Cluttered Push to Target	96.7	100.0	0.402	0.169	0.233	0.185
Pick Eraser from Drawer	93.3	86.7	0.481	0.344	0.137	0.104
Pick Fruit	100.0	100.0	0.541	0.297	0.244	0.188
Stack Cube	93.3	100.0	0.504	0.316	0.188	0.140
Overall	96.7	97.3	0.479	0.283	0.196	0.151

well-structured even under real-world variation. Figure 4 further shows that the compatibility score responds sharply to both noise and temporal perturbations. On RoboTwin-Clean, replacing the conditioning action with a fully random one drives the score from 0.91 to near zero (-0.007); a temporal shift of only ± 2 frames reduces it from 0.91 to 0.64, and a shift of ± 8 frames to 0.42.

4.3 VA-CLIP as a Reference-Free Verifier

We next evaluate whether VA-CLIP can serve as a reference-free verifier for action-conditioned imagined transitions. We study this question from two complementary angles: counterfactual discrimination evaluates whether VA-CLIP separates visually similar transitions with different execution outcomes, while score-reference similarity analysis measures whether its reference-free score correlates with a reference-based signal, i.e., the DINOv2 similarity [41] between ground-truth and Ctrl-World [13] replayed trajectories.

Evaluation protocol. For counterfactual discrimination, we construct three-way groups for each task. Each task contains 10 scenario groups covering lighting and position generalization, and each group includes three visually similar trajectories with different execution outcomes. We treat matched action-video pairs as positives and cross-pairs within the same group as counterfactual mismatches. We report action-to-video and video-to-action ranking accuracy, together with positive scores, counterfactual scores, score gaps, and hard-negative margins; the random baseline is 33.3%.

For score-reference similarity analysis, we train Ctrl-World on 400 collected real-robot trajectories and replay it on 200 held-out validation trajectories. We then sample transition chunks from the replayed rollouts and compute the Spearman correlation between the VA-CLIP compatibility score and the DINOv2 similarity between the corresponding ground-truth and replayed chunks. This reference-based signal is used only for diagnostic analysis, while VA-CLIP itself remains reference-free at evaluation time.

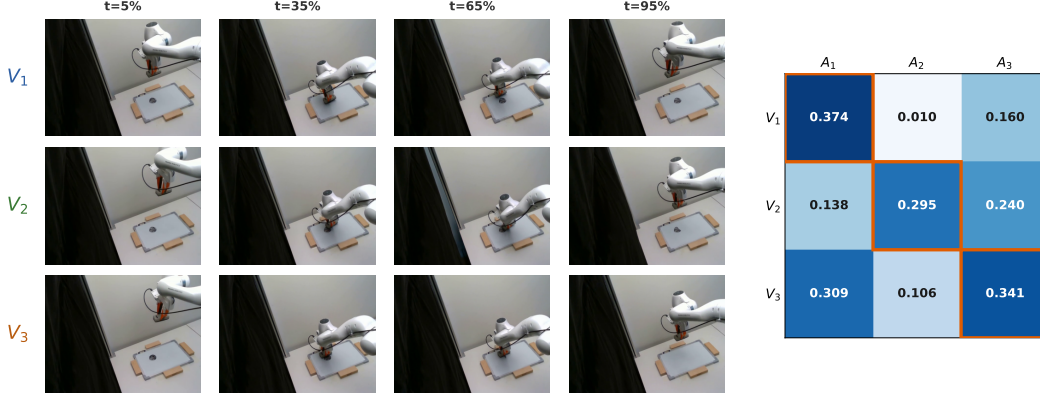


Figure 5: **Counterfactual discrimination for action-consistency verification.** Each group contains three visually similar transitions (V_1 – V_3) from similar initial states but different action-conditioned outcomes, paired with their actions (A_1 – A_3). The 3×3 matrix reports VA-CLIP scores S_{ij} between action A_i and transition V_j . Higher diagonal scores indicate that VA-CLIP selects the true action–transition pairs over visually plausible counterfactual mismatches, supporting reference-free detection of action-inconsistent visual transitions.

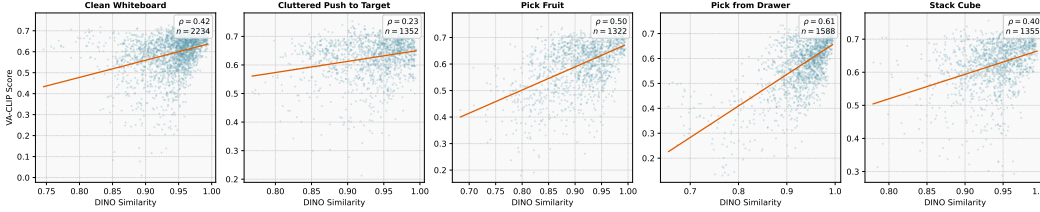


Figure 6: **VA-CLIP score correlates with reference-based rollout fidelity.** Each point denotes one replayed transition chunk, and the orange line shows the OLS trend. VA-CLIP scores are positively correlated with the DINOv2 similarity between ground-truth and replayed chunks, suggesting that higher compatibility scores correspond to more faithful imagined transitions despite requiring no reference future at evaluation time.

Results. As shown in Table 2 and Fig. 5, VA-CLIP achieves strong counterfactual discrimination across all five tasks, with 96.7% action-to-video and 97.3% video-to-action ranking accuracy, far above the 33.3% random baseline. The score statistics also show a clear separation between matched and counterfactual pairs, with positive pairs scoring 0.479 on average and counterfactual mismatches scoring 0.283, resulting in a mean gap of 0.196. These results indicate that VA-CLIP can distinguish the correct action–transition correspondence even among counterfactual alternatives.

Figure 6 provides complementary evidence by comparing the VA-CLIP score with the DINOv2 [29] similarity between ground-truth and replayed trajectories. Across five tasks, the correlation is consistently positive, with Spearman’s ρ ranging from 0.23 to 0.61 and an average of 0.43.

4.4 Improving World-Action Models via Test-Time Selection

Finally, we evaluate whether VA-CLIP can be used beyond offline analysis to improve downstream world-action models.

Evaluation protocol. We conduct experiments on the RoboTwin [28] clean setting with two FastWAM [22] variants: **FastWAM-Joint** and **FastWAM-IDM**. Each FastWAM variant is trained on the Aloha-AgileX embodiment using 50 demo_clean demonstrations per task for 10 epochs. At evaluation time, each task is tested on 50 episodes, and we report the average success rate. For each FastWAM variant, we compare the original model with a VA-CLIP-guided counterpart that reranks

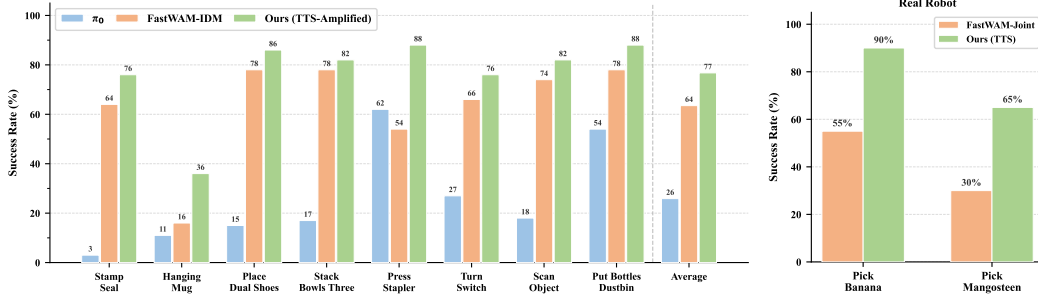


Figure 7: **VA-CLIP-guided test-time selection results.** *Left:* Per-task success rates for π_0 , FastWAM-IDM, and our TTS-amplified variant across eight hard robotwin tasks, together with their average. *Right:* Real-robot evaluation on two pick tasks (20 trials each) comparing FastWAM-Joint against our TTS-amplified variant.

candidate rollouts at test time. We also include π_0 [42] as a policy baseline trained and evaluated under the same data and evaluation setting.

Results. As shown in Fig. 7, VA-CLIP-guided test-time selection yields consistent gains over the base models across all eight hard tasks, with an average improvement of +13%. The gains scale with the number of candidates N : most tasks show a monotonic increase, with the largest absolute benefit at $N = 16$. These results suggest that VA-CLIP provides a useful inference-time signal for selecting more action-compatible imagined transitions, and that the benefit scales with candidate diversity. On real-robot tasks, VA-CLIP-guided selection yields +35% on Pick Banana (55% $\rightarrow$ 90%) and +35% on Pick Mangosteen (30% $\rightarrow$ 65%), demonstrating that the reference-free compatibility signal transfers to physical robot execution.

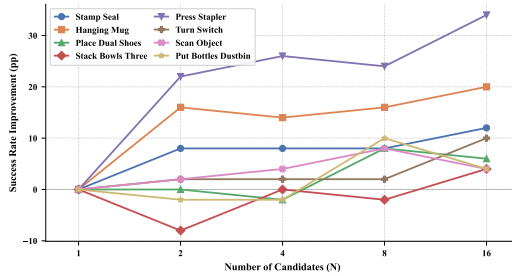


Figure 8: **Per-task improvement over the number of candidates N on the eight hard tasks.**

5 Limitations

VA-CLIP verifies action consistency through a global compatibility score between an action chunk and a visual transition, but it does not yet provide fine-grained semantic guidance about which object, region, or temporal segment causes the mismatch. As a result, our current use is limited to test-time rollout selection rather than directly supervising or correcting world-model training. Future work could develop more localized compatibility signals that provide actionable feedback for improving WAM generation. In addition, our real-robot data contains limited camera deviation. Studying whether stable action–video correlations can be learned under stronger viewpoint changes and more diverse robot embodiments remains an important direction.

6 Conclusion

We introduced VA-CLIP, a video-action contrastive framework for reference-free verification of action-conditioned world-model rollouts. VA-CLIP learns a compatibility score, which identifies action-consistent imagined futures without requiring a ground-truth reference. Across simulated and real-robot data, our results show VA-CLIP captures action-effect correspondence beyond visual similarity and can use this signal to select more action-consistent world-model rollouts at test time.

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